

NIKHIL RAO

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PROFILE

- Associate Software Engineer with experience in **backend and product engineering** for enterprise SaaS platforms, working on business-critical workflows in the insurance domain.
- Strong development background across **Python, PL/SQL, and full-stack web technologies**, with hands-on experience building scalable systems and internal tooling.
- Built and deployed an end-to-end **AI-driven application** in collaboration with the National Institute of Oceanography, applying engineering principles to real-world data and scale.

TECHNICAL SKILLS

- Programming Languages:** Python, PL/SQL, Java, C/C++, JavaScript
- Frameworks:** Django, Next.js, React
- Databases:** MongoDB, MySQL, Oracle Database
- AI / Computer Vision:** PyTorch, YOLOv9, OpenCV

EXPERIENCE

OneShield India Pvt Ltd

July 2025 – Present

Associate Engineer – Product Engineering

- Developed and optimized backend and frontend components for OneShield Enterprise (OSE), a large-scale SaaS platform used by Property & Casualty insurers and MGAs.
- Optimized **policy lifecycle workflows** by tuning queries and introducing appropriate **database indexing**, reducing unnecessary table scans and improving execution performance by approximately **30%**.
- Developed and maintained **PL/SQL-based components** within a complex transactional system, contributing to more efficient data access and processing.
- Built and automated regression test scenarios covering nearly **50% of critical workflows**, improving release confidence and reducing manual validation effort by approximately **12 developer hours per week**.

OneShield India Pvt Ltd

July 2024 – August 2024

Software Engineering Intern

- Completed a structured engineering internship as part of OneShield's onboarding and training program, gaining hands-on experience with **OneShield's proprietary tools and frameworks** and exposure to **SDLC practices, release cycles, and quality gates** within an enterprise SaaS environment.
- Assisted in developing and enhancing frontend components of the **Policy module**, supporting end-to-end workflows such as policy creation, endorsements, and renewals.
- Automated **30+ test scenarios** using **Java, Cucumber, and Gherkin**, translating manual test cases into maintainable automation scripts integrated with CI execution.

KEY PROJECT

Underwater Coral & Sponge Analysis Platform

July 2023 – June 2024

National Institute of Oceanography (NIO)

- Designed and developed a **scalable computer vision system** for automated underwater image analysis used in marine research.
- Trained and evaluated a **YOLOv9-based detection model** on 5000+ images, achieving up to **92% accuracy**.
- Built and integrated a **custom OpenCV-based preprocessing pipeline** for low-visibility underwater images, improving detection robustness across varying water clarity levels.
- Implemented backend services in **Django** to handle asynchronous inference and batch processing of **1000+ images per request**.
- Built a **Next.js frontend** for bulk uploads, result visualization, and workflow management, with **MongoDB** used for scalable storage of metadata, inference results, and experiment tracking.

EDUCATION

Don Bosco College of Engineering

2021 – 2025

B.E in Computer Engineering (Honours in Artificial Intelligence & Machine Learning)

CGPA: 8.82